Name: _____

TOPIC TEST 1

Chapter 3: Arithmetic and geometric sequences

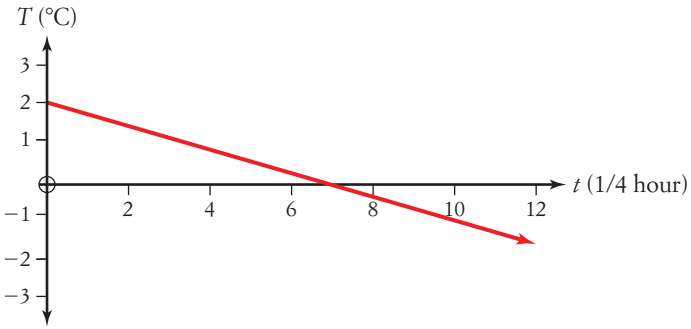
- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 9 short-answer questions (28 marks)
- Part C: 5 analysis questions (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 At the beginning of a particular day in winter in New York the temperature was 2°C at 8 a.m. It became steadily colder during the day, decreasing 0.3°C every quarter of an hour. The temperatures over a period of 3 hours are shown below.



After how many hours, to the nearest hour, is the temperature 0°C?

A 2

B 4

C 5

D 7

E 1

[1 mark]

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2 The first four terms of a sequence are $-2, -8, -14, -20$. The eighth term of the sequence is:

A -2

B -26

C -44

D -50

E -184

[1 mark]

3 The first four terms of a sequence are $-2, -8, -14, -20$. The sum to the eighth term of the sequence is:

A -2

B -26

C -44

D -50

E -184

[1 mark]

4 The common difference $d = t_n - t_{n-1}$ for the sequence $-2, -8, -14, -20$ is:

A -2

B 2

C 6

D -6

E -44

[1 mark]

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5 The common ratio $r = \frac{t_n}{t_{n-1}}$ for the sequence $\frac{1}{2}, -4, 32, -256, \ldots$ is:

A 9

B -9

C 8

D -8

E $\frac{1}{2}$

[1 mark]

Part B

Short-answer questions

9 short-answer questions
28 marks
Show your working where appropriate.

1 Find the arithmetic sequence if the first term in the sequence is 5 and the 10th term is -10.

[3 marks]

2 If the first six terms of a sequence are $-2, -5, -8, -11, -14, -17$, state the sum of the sequence to 10 terms.

[3 marks]

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3 Find the sum of the arithmetic sequence 7, 14, 21, 35, …, 112.

[4 marks]

4 Calculate the sum of all the multiples of 5 between 1 and 51.

[4 marks]

5 If x, $x - 1$, and $x + 3$ are the first three terms of a geometric sequence, calculate the value of x.

[2 marks]

6 Find the first four terms of the sequence described by the recursion:

$$a_1 = 5 \text{ and } a_n = -2a_{n-1} + 2.$$

[2 marks]

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7 If the first six terms of a sequence are 2, 6, 18, 243, 729, 2187, state the value of the 12th term of the sequence.

[3 marks]

8 Find the common ratio $r = \frac{t_n}{t_{n-1}}$ for the sequence $\frac{2}{3}, -\frac{4}{9}, \frac{8}{27}, -\frac{16}{81}, \ldots$

[2 marks]

9 Find the geometric sequence if the fourth term in that sequence is -2 and the 10th term is -32.

[5 marks]

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Part C

Analysis questions

5 analysis questions
18 marks
Show your working where appropriate.

1 The price of shares in a particular company is falling by an average of 2% each day.

a What percentage of their value are the shares worth after 2 days?

b How many days will it take for the shares to halve in value?

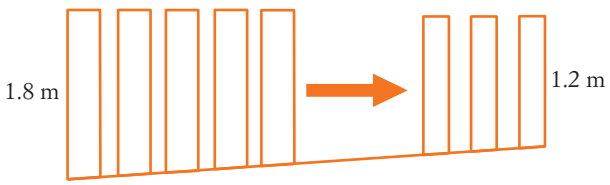
c Find the value of the shares after 4 weeks if they were originally worth $9.55.

[4 marks]

2 A geometric series has a limiting sum of 40 and its first term is 5. Find the common ratio and hence the geometric sequence.

[3 marks]

3 A timber fence is to be built on sloping land, with the shortest piece of timber 1.2 m and the longest piece 1.8 m. There are 61 pieces of timber in the fence.



a What is the difference in height between each piece of timber?

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b What length of timber is needed for the fence altogether?

[3 marks]

4 A tree 3 m high grows another $2\frac{2}{5}$ m after 5 years, then another $1\frac{23}{25}$ m after 5 more years and so on, each year growing $\frac{4}{5}$ more than the previous year's growth. Find the eventual height of the tree.

[3 marks]

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5 A ball is dropped from a height of 2 metres and bounces up to $\frac{2}{5}$ of its original height. It continues bouncing, rising $\frac{2}{5}$ of its height on each bounce until it eventually reaches the ground.

a What is the total distance through which it travels?

b Will the ball ever stop bouncing?

[5 marks]

[Total marks: 51]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 From the graph, $T = 0$ when $t = 7$.

No. of hours $= \frac{1}{4} \times 7 = 1\frac{3}{4} \approx 2$.

$\therefore$ **A**

2 $a = -2$

$d = -8 - (-2) = -6$

$t_n = -2 - 6(n - 1)$

$t_{10} = -2 - 6(8 - 1)$

$= -2 - 6 \times 7$

$= -44$

$\therefore$ **C**

3 $a = 2$

$d = -6$

Use the formula:

$S_n = \frac{n}{2}[2a + (n - 1)d]$

$S_8 = \frac{8}{2}[-4 + 7 \times -6]$

$= 4(-4 - 42)$

$= -184$

$\therefore$ **E**

4 $d = t_n - t_{n-1}$

$= -8 - (-2)$

$= -14 - (-8)$

$= -20 - (-14)$

$= -6$

$\therefore$ **D**

5 $r = \frac{t_n}{t_{n-1}}$

$= \frac{-4}{\frac{1}{2}} = \frac{32}{-4} = \frac{-256}{32}$

$= -8$

$\therefore$ **D**

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Part B – Short-answer questions

1 $t_1 = a = 5$ and $t_{10} = -10$

Use $a = 5$ and $t_n = a + (n - 1)d$ to solve for d.

$5 + (10 - 1)d = -10$ **[1 mark]**

$9d = -15$

$d = -\frac{5}{3}$ **[1 mark]**

The arithmetic sequence is $5, \frac{10}{3}, \frac{5}{3}, 0, \ldots$ **[1 mark]**

2 For the sequence $-2, -5, -8, -11, -14, -17$, $a = -2$, $d = -3$. **[2 marks]**

Use $S_n = \frac{n}{2}[2a + (n - 1)d]$ to find S_{10}.

$S_{10} = \frac{10}{2}[2(-2) + (10 - 1)(-3)]$

$\therefore S_{10} = -155$ **[1 mark]**

3 For the arithmetic sequence $7, 14, 21, 35, \ldots, 112$, $a = 7$, $d = 7$, $l = 112$. **[2 marks]**

Use $S_n = \frac{n}{2}(a + l)$ where $l = t_n = a + (n - 1)d$.

Solve for n, $l = t_n = 112 = 7 + (n - 1)7$ gives $n = 16$. **[1 mark]**

$\therefore S_{16} = \frac{16}{2}(7 + 112)$ **[1 mark]**

$\therefore S_{16} = 952$

4 Arithmetic sequence is $5, 10, 15, 20, \ldots, 50$, where $a = 5$, $d = 5$, $l = 50$.

[2 marks]

We need to choose 50 because that is the last multiple of 5 before 51.

Use $S_n = \frac{n}{2}(a + l)$ where $l = t_n = a + (n - 1)d$.

Solve for n, $l = t_n = 50 = 5 + (n - 1)5$ gives $n = 10$. **[1 mark]**

$\therefore S_{10} = \frac{10}{2}(5 + 50)$ **[1 mark]**

$\therefore S_{10} = 275$

The other sum formula could also be used, using $n = 10$.

5 Using $r = \frac{t_2}{t_1} = \frac{t_3}{t_2} = \frac{t_4}{t_3} = \ldots$ gives $r = \frac{x - 1}{x} = \frac{x + 3}{x - 1}$. **[1 mark]**

$x(x + 3) = (x - 1)(x - 1)$

$x^2 + 3x = x^2 - 2x + 1$

$0 = -5x + 1$

$5x = 1$

$x = \frac{1}{5}$ **[1 mark]**

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6 Sequence is 5, −8, 18, −34. **[2 marks]**

7 Sequence 2, 6, 18, 243, 729, 2187, where $a = 2, r = 3$. **[2 marks]**

Using $t_n = ar^{n-1}$ to find t_{12}

$t_{12} = 2(3)^{12-1}$ **[1 mark]**

$\therefore t_{12} = 354\,294$

8 Using $r = \dfrac{t_n}{t_{n-1}}$ to find $r = \dfrac{-\frac{4}{9}}{\frac{2}{3}}$ **[1 mark]**

Gives $r = -\dfrac{2}{3}$. **[1 mark]**

9 $t_4 = -2$ and $t_{10} = -32$ **[1 mark]**

$t_n = ar^{n-1}$ to solve simultaneously for a and r.

$ar^{4-1} = -2$ and $ar^{10-1} = -32$ **[1 mark]**

gives $a = -\dfrac{1}{2}, r = \sqrt[3]{4}$ or $a = \dfrac{1}{2}, r = -\sqrt[3]{4}$ **[2 marks]**

The geometric sequence is $-\dfrac{1}{2}, -\dfrac{\sqrt[3]{4}}{2}, -\dfrac{\sqrt[3]{4^2}}{2}, \ldots$ or $\dfrac{1}{2}, -\dfrac{\sqrt[3]{4}}{2}, \dfrac{\sqrt[3]{4^2}}{2}, \ldots$ **[1 mark]**

Part C – Analysis questions

1 a Falling by 2% means the shares are worth 98% of their original price.

$0.98 \times 0.98 = 0.9604$

$= 96.04\%$ **[2 marks]**

b $0.98^{n-1} = 0.5$

Use solve on calculator = 35 days **[1 mark]**

c $9.55 \times 0.98^{16} = \$6.91$ **[1 mark]**

2 Using $S_\infty = \dfrac{a}{1-r}$ for a limiting sum of 40 where $a = 5$. **[1 mark]**

Solve $40 = \dfrac{5}{1-r}$

$\therefore r = \dfrac{7}{8}$ **[1 mark]**

Sequence is $5, \dfrac{35}{8}, \dfrac{245}{64}, \dfrac{1715}{512}, \ldots$ **[1 mark]**

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3 a $a = 1.2, l = 1.8, n = 61$ **[1 mark]**

Using $t_n = a + (n - 1)d$ with $l = t_{61} = 1.2 + (61 - 1)d = 1.8$

Gives $d = 0.01$

$\therefore$ Difference in height between each piece of timber is 0.01 m. **[1 mark]**

b Using $S_n = \frac{n}{2}[2a + (n - 1)d]$ where $a = 1.2, d = 0.01, n = 61$

length of timber = 91.5 metres. **[1 mark]**

4 $a = 3, r = \frac{4}{5}$ **[1 mark]**

Use $S_\infty = \frac{a}{1 - r}$ with $a = 3, r = \frac{4}{5}$. **[1 mark]**

$\therefore$ Eventual height of tree = 15 metres. **[1 mark]**

5 a Use $S_\infty = \frac{a}{1 - r}$ with $a = \frac{2}{5}, r = \frac{2}{5}$ **[2 marks]**

Distance travelled is original 2 metres plus twice the height of each subsequent bounce.

Total distance travelled =

$$2 + 2\left(\frac{a}{1 - r}\right)$$

$$= 1 + 2\left(\frac{\frac{2}{5}}{1 - \frac{2}{5}}\right)$$

$$= 1 + 2\left(\frac{\frac{2}{5}}{\frac{3}{5}}\right)$$

$$= 1 + \frac{4}{3}$$

$$= \frac{7}{3}$$

$$= 2\frac{1}{3} \text{ metres}$$ **[2 marks]**

b As this is a limiting sum, in theory, the ball will never stop bouncing.

[1 mark]

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